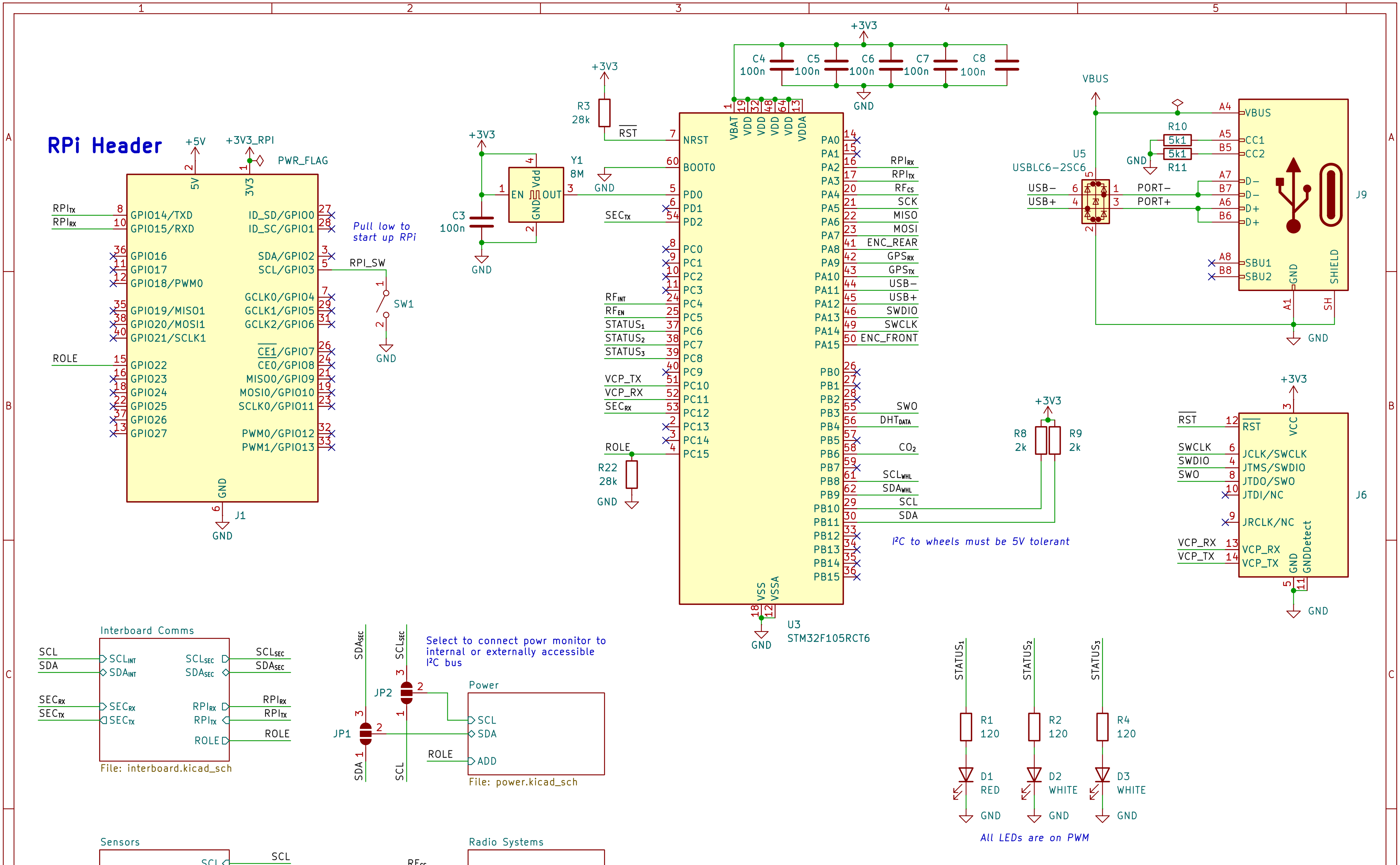


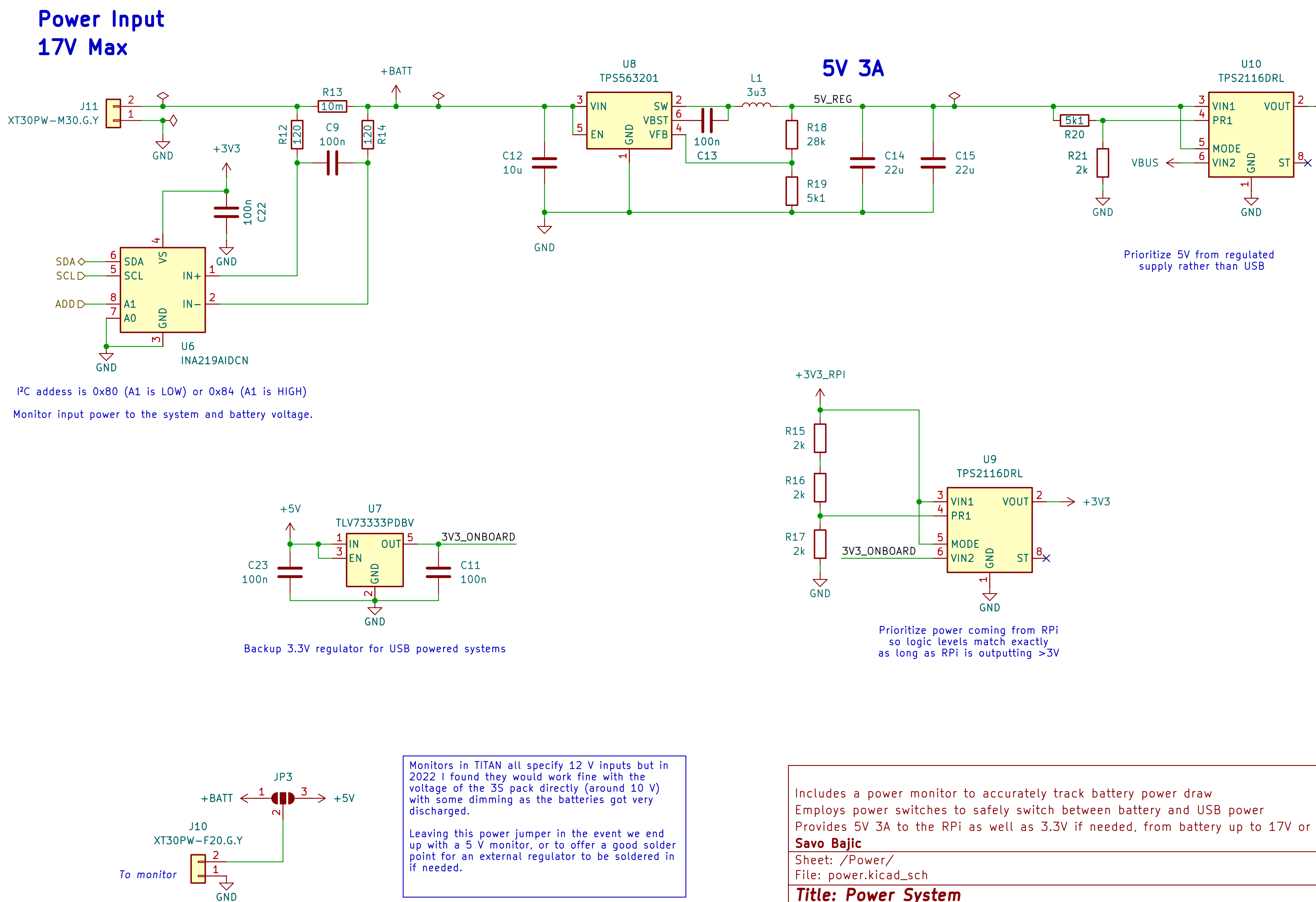
Assume they are the primary unless the board detects a connection to another board
The microcontroller is only really active on the primary board
Central microcontroller collecting and routing data around TITAN

Savo Bajic

Sheet: /
File: titan_2026_rev_a.kicad_sch

Title: TITAN HAT 2026 Core

Size: USLetter	Date: 2026-04-23	Rev: A
KiCad E.D.A. 10.0.5		Id: 1/5



Includes a power monitor to accurately track battery power draw
Employs power switches to safely switch between battery and USB power
Provides 5V 3A to the RPi as well as 3.3V if needed, from battery up to 17V or USB's 5V

Savo Bajic

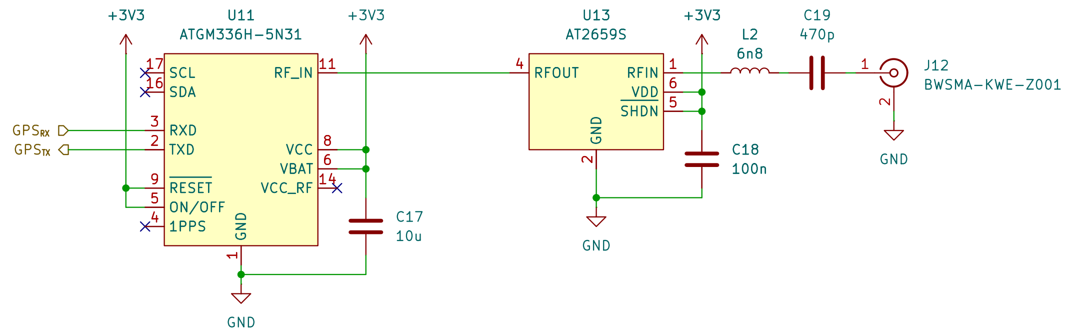
Sheet: /Power/
File: power.kicad_sch

Title: Power System

Size: USLetter	Date: 2026-04-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/5

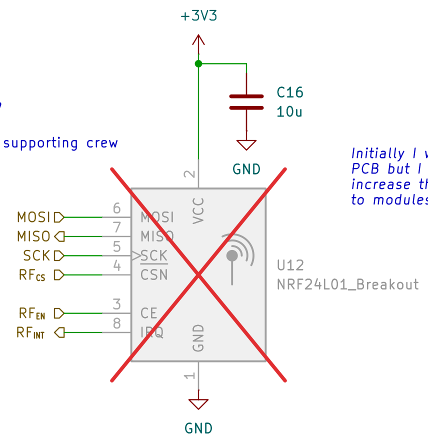
GPS

GPS is used primarily as a backup source for distance and speed information but location can also be recorded during the run.



Telemetry

Broadcast data to supporting crew



Initially I wanted to embed the nRF24 fully into the PCB but I realized that would take a lot of time and increase the PCBA cost a heap so I decided to stick to modules, albeit better placed.

Radio communications between TITAN and the rest of the team as well as GPS

Savo Bajic

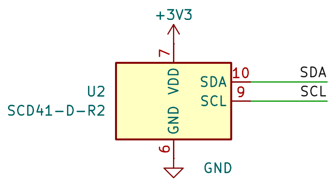
Sheet: /Radio Systems/

File: radio.kicad_sch

Title: Radio Systems

Size: USLetter	Date: 2026-04-22	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/5

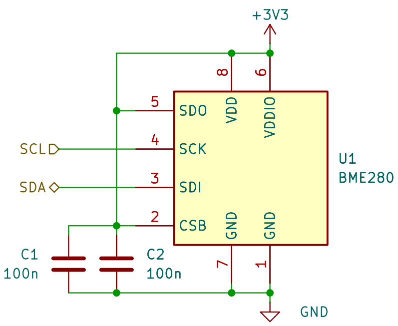
SCD-41



I²C addess is 0x62

Measures CO_2 up to 5000ppm
SCD-40 can be used for up to 2000ppm

BME280



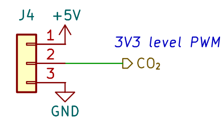
I²C addess is 0x77

Our interest in atmospheric conditions is to derive ambient air conditions for performance calculations, for this temperature, humidity, and ideally pressure too are used.

This sensor provides all three. Locating it on the board may cause misleading readings due to the inside of TITAN heating up in the sun but readings near the start of runs should be usable.

Redundant sensors kept from 2022

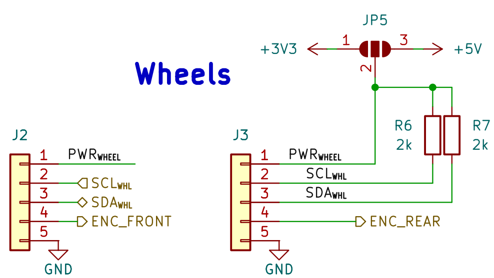
MH-Z19



Digital CO_2 sensor that measures up to 2000ppm using infrared. Outputs result using PWM.

Sensor is placed near the riders' heads.

Wheels

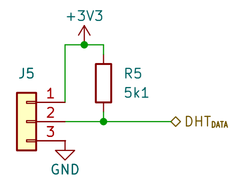


All wheel signals can be 5V!
I²C addesses programmable.
In 2022 front was 0x01 and rear was 0x02.

Both wheels have an MLX90614 sensor to monitor brake disk temeprature, and the ability to have a digital encoder for wheel speed although both encoders may not be used in the end.

In 2022 they were provided 5V but the MLX90614s used were rated for 3.3V, the system worked however so I've included the solder jumper to set the wheel voltage as desired.

DHT11



Temperature and humidity, but far slower to read and update.

Located off board by air intake avoiding high speed inflow.

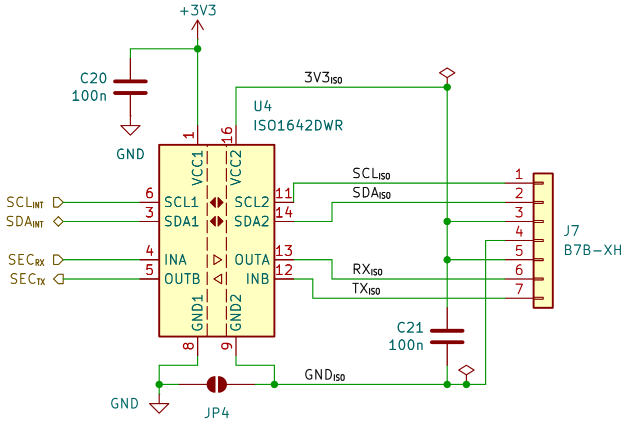
Some sensors are connected from off-board to be better capture data
DHT11 and MH-Z19 connections are kept from 2022, serving as backups onboard sensors

All the sensors for monitoring the bike and the environment
Savo Bajic
Sheet: /Sensors/
File: sensors.kicad_sch

Title: Sensing System

Size: USLetter	Date: 2026-04-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/5

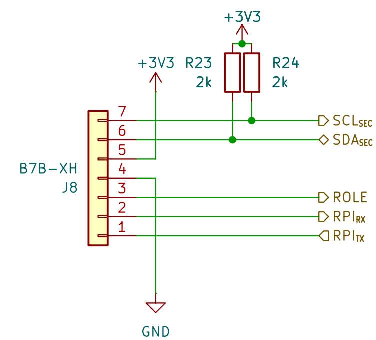
Primary Input
(from Secondary Output)



For safety the primary side has an isolator to separate the systems

Expecting standard crossover cables
hence the reversed numbering

Secondary Out
(to Primary In)



The ROLE signal is used determine if a board is a primary or secondary system via hardware. It is pulled down to GND by default which indicates the board is a "primary" board. Once a bridge is formed between two boards, a loop is closed for the secondary board through the primary board which then pulls the ROLE line high.

This ROLE is used primarily to disable the secondary microcontroller so it doesn't interfere with normal operation as well as assign the secondary system's power monitor a different I²C address to the primary board's sensor. The RPi can read ROLE to determine state as well.

Unfortunately some jumpers are still needed to ensure that the primary board can only reach the secondary power monitor via I²C. If the entire secondary I²C bus was available to it then the other sensors would have address collisions preenting normal his should be used as a startup check in firmware to see if the bridges are done correctly on both units.

On the secondary board, the "secondary output" is used to connect to the primary's input
A bit of a mess to include on the main page, so I moved it here

Savo Bajic

Sheet: /Interboard Comms/
File: interboard.kicad_sch

Title: Interboard Connections

Size: USLetter	Date: 2026-04-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/5